Lecture Notes 2

Econometrics

The Multiple Linear Regression Model

Markeljan Blacëri

Department of Economics and Management
University of Padova
markeljan.blaceri@phd.unipd.it

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Motivation for Multiple Regression

- Consider the following wage equation:

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \varepsilon$$

- Now we are able to measure the effect of education on wage, holding experience fixed.
- In a simple regression, we would have to assume that experience is uncorrelated with education.

Motivation for Multiple Regression

- Suppose family consumption is a quadratic function of family income:

$$cons = \beta_0 + \beta_1 inc + \beta_2 inc^2 + \varepsilon$$

- Consumption depends on only one observed variable, that is, income.
- However, the model falls outside simple regression because it contains two functions of income.
- The marginal propensity to consume is approximated by:

$$\frac{\Delta cons}{\Delta inc} \approx \beta_1 + 2\beta_2 inc$$

Multiple Linear Regression Model

- The general regression model with additive errors:

$$y = f(x_1, x_2, \ldots, x_k) + \varepsilon$$

- A linear regression model is obtained when $f(x_1, x_2, \ldots, x_k)$ is specified to be a linear function:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + \varepsilon$$

where:

- β_0 is the intercept,
- β_1 is the parameter associated with x_1,
- β_2 is the parameter associated with x_2,

  $\vdots$
- β_k is the parameter associated with x_k.

Obtaining the OLS Estimates

The OLS estimates, $k+1$ of them, are chosen to minimize the sum of squared residuals:

$$\sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_{i1} - \ldots - \widehat{\beta}_k x_{ik} \right)^2$$

This minimization problem leads to $k+1$ linear equations in $k+1$ unknowns $\widehat{\beta}_0, \widehat{\beta}_1, \ldots, \widehat{\beta}_k$, the **first-order conditions**:

$$\begin{aligned}
\sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_{i1} - \ldots - \widehat{\beta}_k x_{ik} \right) &= 0 \\
\sum_{i=1}^{n} x_{i1} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_{i1} - \ldots - \widehat{\beta}_k x_{ik} \right) &= 0 \\
&\vdots \\
\sum_{i=1}^{n} x_{ik} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_{i1} - \ldots - \widehat{\beta}_k x_{ik} \right) &= 0
\end{aligned}$$

The Multiple Linear Regression Model in Matrix Form

- Solving these $k+1$ equations by hand is impractical. It is far more convenient to write the model in **matrix form**.
- For n observations, the multiple linear regression model can be written as:

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}_{\mathbf{y}_{n\times 1}} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix}}_{\mathbf{X}_{n\times(k+1)}} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}}_{\boldsymbol{\beta}_{(k+1)\times 1}} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{\boldsymbol{\varepsilon}_{n\times 1}}$$

- Using compact matrix notation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Obtaining the OLS Estimates

- The OLS estimator is defined to be the estimator that minimizes the sum of squared errors:

$$\min_{\boldsymbol{\beta}} S(\boldsymbol{\beta}) = \sum_{i=1}^{n} \varepsilon_i^2 = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

- Expanding the quadratic (using $\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} = \mathbf{y}'\mathbf{X}\boldsymbol{\beta}$, a scalar):

$$S(\boldsymbol{\beta}) = \mathbf{y}'\mathbf{y} - 2\boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$$

- FOC:

$$\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$$

- Setting the FOC equal to $\mathbf{0}$ yields the OLS estimator:

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

Interpreting the OLS Regression Equation

- Consider the case of two independent variables:

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x_1 + \widehat{\beta}_2 x_2$$

- The estimates $\widehat{\beta}_1$ and $\widehat{\beta}_2$ have partial effect, or ceteris paribus, interpretations.

$$\Delta\widehat{y} = \widehat{\beta}_1 \Delta x_1 + \widehat{\beta}_2 \Delta x_2$$

- When x_2 is held fixed, so that $\Delta x_2 = 0$, then

$$\Delta\widehat{y} = \widehat{\beta}_1 \Delta x_1$$

holding x_2 fixed.

- Similarly, holding x_1 fixed

$$\Delta\widehat{y} = \widehat{\beta}_2 x_2$$

Example: Hourly Wage Equation

Assume we want to estimate the following regression model:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \varepsilon$$

Using a data from 526 workers, the estimated equation is:

$$\log(wage) = 0.284 + 0.092 educ + 0.0041 exper + 0.022 tenure$$

Ceteris paribus interpretation:

- The coefficient 0.092 means that, holding *exper* and *tenure* fixed, another year of education is predicted to increase wage by 9.2%.

A Partialling Out Interpretation

Consider $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x_1 + \widehat{\beta}_2 x_2$. The coefficient $\widehat{\beta}_1$ can be recovered in two steps:

1. Regress x_1 on x_2 (and a constant) and keep the residuals $\widehat{r}_{i1}$.
2. Regress y on these residuals alone:

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n} \widehat{r}_{i1}\, y_i}{\sum_{i=1}^{n} \widehat{r}_{i1}^2}.$$

- The residuals $\widehat{r}_{i1}$ are the part of x_1 left *after partialling out* x_2: the variation in x_1 uncorrelated with x_2.
- Hence $\widehat{\beta}_1$ measures the effect of x_1 on y once the influence of x_2 has been netted out — the precise sense of "holding x_2 fixed."
- This generalizes (Frisch–Waugh–Lovell): each $\widehat{\beta}_j$ is a simple regression of y on the part of x_j orthogonal to all other regressors.

Sums of Squares

To assess how well the estimated regression fits the data, we can decompose the total variation in y into two components.

- **Explained Sum of Squares (ESS):**

$$ESS = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- **Residual Sum of Squares (RSS):**

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_{i=1}^{n} \hat{\varepsilon}_i^2$$

- **Total Sum of Squares (TSS):**

$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2 = ESS + RSS$$

Goodness-of-Fit

How well does the estimated regression line fit the observations?

- A popular measure for the goodness-of-fit of a regression model is the proportion of the (sample) variance of y that is explained by the model.
- This variable is called the R^2 and is defined as:

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$$

- R^2 will never decrease if the number of regressors is increased, even if the additional variables have no real explanatory power.
- Adjusted R^2:

$$\overline{R}^2 = 1 - \frac{\frac{1}{N-K} RSS}{\frac{1}{N-1} TSS}$$

Example: Hourly Wage Equation

Using a data from 526 workers, the estimated equation is:

$$\log(wage) = 0.284 + 0.092educ + 0.0041exper + 0.022tenure$$

In addition, the sums of squares are:

$$ESS = 46.87, \quad RSS = 101.46, \text{ and } TSS = 148.33$$

Zero Conditional Mean

Zero Conditional Mean Assumption

The error ε has an expected value of zero given any values of the independent variables.

$$\mathbb{E}(\varepsilon|\mathbf{X}) = 0$$

- This assumption implies that ε is uncorrelated with each explanatory variable.

$$\text{Cov}(x_{ij}, \varepsilon_i) = 0 \quad \forall j = 1, \ldots, k$$

- When the zero conditional mean assumption holds, we often say that we have exogenous explanatory variables.
- The assumption fails if ε includes an omitted variable correlated with any of the x_j.

Unbiasedness of the OLS Estimator

The OLS estimator $\widehat{\boldsymbol{\beta}}$ is unbiased meaning that, in repeated sampling, we can expect that the OLS estimator is on average equal to the true value $\boldsymbol{\beta}$.

$$\begin{aligned}\mathbb{E}\left[\widehat{\boldsymbol{\beta}}\right] &= \mathbb{E}\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}\right] = \mathbb{E}\left[\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})\mathbf{X}'\boldsymbol{\varepsilon}\right] \\ &= \boldsymbol{\beta} + \mathbb{E}\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\right] = \boldsymbol{\beta} + \mathbb{E}\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right]\underbrace{\mathbb{E}[\boldsymbol{\varepsilon}]}_{=\mathbf{0}} \\ &= \boldsymbol{\beta}\end{aligned}$$

The OLS estimator is unbiased as long as the error terms are mean zero and independent of all explanatory variables.

Variance of the OLS Estimator

- To obtain the variance of the OLS estimators, we add a homoskedasticity assumption for two reasons:
  1. The formulas are simplified.
  2. OLS has an important efficiency property.

Homoskedasticity

The error ε has the same variance given any value of the explanatory variables.

$$\text{Var}(\varepsilon|\mathbf{X}) = \sigma^2$$

Variance of the OLS Estimator

Just as in simple regression, homoskedasticity and no serial correlation give $\mathbb{E}(\varepsilon\varepsilon' \mid \mathbf{X}) = \sigma^2 \mathbb{I}_n$. The variance of $\widehat{\boldsymbol{\beta}}$ (conditional on $\mathbf{X}$) is then

$$\begin{aligned}
\text{Var}\left[\widehat{\boldsymbol{\beta}}\right] &= \mathbb{E}\left[\left(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)\left(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)'\right] = \mathbb{E}\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\right] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\left(\sigma^2\mathbb{I}_n\right)\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}
\end{aligned}$$

The $(k+1) \times (k+1)$ matrix $\text{Var}\left[\widehat{\boldsymbol{\beta}}\right]$ is a variance-covariance matrix: it contains the variances of $\widehat{\beta}_0, \widehat{\beta}_1, \ldots, \widehat{\beta}_k$ on the diagonal, and their covariances as off-diagonal elements.

Variance of the OLS Estimator

Estimated Variances and Standard Errors

- As before, these formulas are not usable directly, since σ^2 is unknown. We replace it with an unbiased estimator:

$$\widehat{\sigma^2} = \frac{1}{n-k-1}\sum_{i=1}^{n}\widehat{\varepsilon}_i^2 = \frac{RSS}{n-k-1}$$

- The only change from simple regression is the degrees of freedom: $n-2$ becomes $n-k-1$, one for each of the $k+1$ estimated parameters. Its square root $\widehat{\sigma}$ is the **standard error of the regression**.
- The estimated variance-covariance matrix is then

$$\widehat{\mathrm{Var}}(\widehat{\boldsymbol{\beta}}) = \widehat{\sigma^2}(\mathbf{X}'\mathbf{X})^{-1}$$

- The **standard error** of $\widehat{\beta}_j$ is the square root of the j-th diagonal element:

$$\mathrm{se}(\widehat{\beta}_j) = \widehat{\sigma}\sqrt{[(\mathbf{X}'\mathbf{X})^{-1}]_{jj}}$$

Including Irrelevant Variables in a Regression Model

Suppose we include a regressor x_j whose true coefficient is zero ($\beta_j = 0$) — the model is *overspecified*.

- **No bias.** OLS remains unbiased for every coefficient: unbiasedness only requires the zero conditional mean assumption, which still holds.

$$\mathbb{E}(\widehat{\beta}_j) = 0, \qquad \mathbb{E}(\widehat{\beta}_l) = \beta_l \text{ for the others.}$$

- **But a variance cost.** Including the irrelevant variable can raise R_l^2 for the others, inflating $\text{Var}(\widehat{\beta}_l)$ and reducing precision.
- **Lesson:** adding junk regressors does not bias OLS, but it wastes degrees of freedom and can hurt efficiency.

Omitting a Relevant Variable

The mirror image of including an *irrelevant* variable: what if we *leave out* one that belongs? Suppose the true model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon,$$

but we estimate the short regression $\widetilde{y} = \widetilde{\beta}_0 + \widetilde{\beta}_1 x_1$, dropping x_2. Then

$$\mathbb{E}(\widetilde{\beta}_1) = \beta_1 + \beta_2\,\delta_1, \qquad \delta_1 = \frac{\mathrm{Cov}(x_1, x_2)}{\mathrm{Var}(x_1)},$$

where δ_1 is the slope of x_2 regressed on x_1. The **omitted variable bias** is $\beta_2\,\delta_1$.

- The bias vanishes only if $\beta_2 = 0$ (x_2 truly irrelevant) or $\delta_1 = 0$ (x_1, x_2 uncorrelated).

The Direction of Omitted Variable Bias

The sign of the bias $\beta_2 \, \delta_1$ depends on the sign of β_2 and the correlation between x_1 and x_2:

	$\text{Corr}(x_1, x_2) > 0$	$\text{Corr}(x_1, x_2) < 0$
$\beta_2 > 0$	positive bias (upward)	negative bias (downward)
$\beta_2 < 0$	negative bias (downward)	positive bias (upward)

Example (ability bias). In the return to education, ability has $\beta_2 > 0$ and is positively correlated with schooling, so omitting it biases the estimated return *upward* — exactly the story from the simple-regression lecture.

Inclusion of Bad Controls

Not every additional control improves a regression. A **bad control** is a variable that is itself an *outcome* of the regressor of interest.

- Adding controls is usually thought to reduce omitted-variable bias — but controlling for a *consequence* of x can instead *create* bias.
- **Rule of thumb:** good controls are determined *before* the regressor of interest (pre-treatment); variables realized *after* it are suspect.

The Angrist–Pischke example. We want the effect of college graduation c on earnings y. Let w be a white-collar-job dummy. Consider two regressions:

$$y = \alpha_0 + \alpha_1 c + u \qquad \text{(college only)}$$
$$y = \gamma_0 + \gamma_1 c + \gamma_2 w + v \qquad \text{(adding occupation as a control)}$$

Which coefficient answers our question — α_1 or γ_1?

Inclusion of Bad Controls

Why Conditioning on Occupation Fails

The problem: occupation w is not a pre-existing characteristic — it is *itself an outcome of* college. Graduating changes both earnings *and* the kind of job a person holds.

- College raises earnings through two channels: directly (higher pay in a given job) and indirectly (by making a white-collar job more likely).
- α_1 captures the full effect — both channels.
- γ_1 holds w fixed, comparing graduates and non-graduates *within the same occupation*. This shuts down the indirect channel, so γ_1 answers a different question.

Worse, the comparison within $w = 1$ is not clean: graduates and non-graduates who both ended up in white-collar jobs are *selected differently* (a non-graduate needs more of everything *else* to get there). Controlling for w thus introduces **selection bias** into γ_1.

A variable realized after the regressor of interest is a bad control.

Regression Through the Origin

Sometimes theory forces the line through the origin, i.e. we impose $\beta_0 = 0$:

$$\widetilde{y} = \widetilde{\beta}_1 x_1 + \widetilde{\beta}_2 x_2 + \ldots + \widetilde{\beta}_k x_k.$$

In the single-regressor case the estimator is

$$\widetilde{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}.$$

- If the true intercept is nonzero, dropping it makes all slope estimates **biased**.
- The residuals no longer sum to zero, and R^2 can even be negative — the usual decomposition breaks down.
- Only impose it when a zero intercept is genuinely justified by theory.

Models with Quadratics

- Quadratic functions are also used quite often in applied economics to capture decreasing or increasing marginal effects.
- In the simplest case, y depends on a single observed factor x, but it does so in a quadratic fashion:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$$

- Does β_1 measure the change in y with respect to x? **NO!**
- If we write the estimated equation as

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x + \widehat{\beta}_2 x^2$$

then we have the approximation

$$\Delta\widehat{y} \approx \left(\widehat{\beta}_1 + 2\widehat{\beta}_2 x\right)\Delta x, \text{ so } \Delta\widehat{y}/\Delta x \approx \widehat{\beta}_1 + 2\widehat{\beta}_2 x$$

Example: Wage and Experience

Consider the estimated regression

$$\widehat{wage} = 3.73 + 0.298\, exper - 0.0061\, exper^2.$$

- The positive *exper* and negative $exper^2$ coefficients imply a **diminishing** return to experience.
- The marginal effect is $\dfrac{\Delta \widehat{wage}}{\Delta exper} \approx 0.298 - 2(0.0061)\, exper$.
- It hits zero at $exper^* = \dfrac{0.298}{2(0.0061)} \approx 24.4$ years — beyond which extra experience is predicted to *lower* wage.

Models with Interaction Terms

- Sometimes, it is natural for the partial effect, elasticity, or semi-elasticity of the dependent variable with respect to an explanatory variable to depend on the magnitude of yet another explanatory variable.
- For example, in the model

$$price = \beta_0 + \beta_1 sqrft + \beta_2 bdrms + \beta_3 sqrtf \cdot bdrms + \beta_4 bthrms + \varepsilon$$

the partial effect of *bdrms* on *price* (holding all other variables fixed) is

$$\frac{\Delta price}{\Delta bdrms} = \beta_2 + \beta_3 sqrft$$

- If $\beta_3 > 0$, then this implies that an additional bedroom yields a higher increase in housing price for larger houses.

Multicollinearity

Multicollinearity arises when two or more explanatory variables are highly (but not perfectly) linearly related.

- **Intuition.** OLS estimates β_j by using the variation in x_j that is *left over* after removing its relationship with the other regressors. If x_j moves almost in lockstep with another variable, little independent variation remains — and the effect of x_j is hard to pin down.
- This inflates the variance of the estimate:

$$\text{Var}(\widehat{\beta}_j) = \frac{\sigma^2}{SST_j\,(1 - R_j^2)}, \qquad \text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is from regressing x_j on *all other* regressors. As $R_j^2 \to 1$, the variance explodes.

The Gauss-Markov Theorem

Add the homoskedasticity assumption $\text{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2 \mathbb{I}_n$ to the earlier ones (linearity, no perfect collinearity, zero conditional mean).

Gauss-Markov Theorem

Under these assumptions, OLS is **BLUE**: the **B**est **L**inear **U**nbiased **E**stimator.

- Among all estimators that are linear in $\mathbf{y}$ and unbiased, OLS has the smallest variance.
- Formally, for any other linear unbiased estimator $\widetilde{\boldsymbol{\beta}}$,

$$\text{Var}(\widetilde{\boldsymbol{\beta}}) - \text{Var}(\widehat{\boldsymbol{\beta}}) \text{ is positive semidefinite.}$$

- So no linear unbiased estimator delivers a smaller variance for any coefficient, or any linear combination of them.

References

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